In this report we use a deterministic "Ross-Macdonald" type spatial kernel model to describe the spread of infection between livestock (hosts) and mosquitoes (vectors) within and between different geographical locations during an outbreak.
Q1a: Consider the following equations:

$$\frac{dL_v}{dt} = B_v - \theta L_v - \frac{L_v^2}{C}, \qquad \frac{dN_v}{dt} = \theta L_v - \mu_v N_V.$$

To determine the Larval population size at disease-free equilibrium, $L_v^\star$, we set both $\frac{dL_v}{dt}$ and $\frac{dN_v}{dt}$ to zero yielding

$$B_v - \theta L_v^* - \frac{(L^*)_v^2}{C} = 0, \tag{1}$$
$$\theta L_v^* - \mu_v N_V^* = 0. \tag{2}$$

Using (4) to solve for L_v^*, we get $L_v^* = \frac{\mu_v N_v^*}{\theta}$. Then, we can give an expression for B_v by substituting this result into (3) and also assuming $C = N_v^*$. This yields

$$B_v = \mu_v N_v^* + \frac{\mu_v^2 (N_v^*)^2}{\theta^2 C} = \left(1 + \frac{\mu_v}{\theta^2}\right) \cdot \mu_v N_v^*.$$

Q1b: Assuming an application of vector control corresponds to changing C into $\tilde{C}$ and results in a relative ATD of $\tilde{n}_v$, once the system has reached its new equilibrium, we can determine $\tilde{C}$ in terms of $N_v^*, \tilde{n}_v, \theta$ and μ_v as follows: We know

$$\tilde{n}_v = \frac{\tilde{N}_v^*}{N_v^*}, \text{ or equivalently, } \quad \tilde{N}_v^* = \tilde{n}_v N_v^*.$$

Therefore, by substituting into our laraval populsation size at disease-free equilibrium, we have

$$\tilde{L}_v^* = \frac{\mu_v \tilde{n}_v N_v^*}{\theta}. \tag{3}$$

Then, substituting this result into (3), we have

$$B_v - \theta \cdot \frac{\mu_v \tilde{n}_v N_v^*}{\theta} - \frac{\mu_v^2 \tilde{n}_v^2 (N_v^*)^2}{\theta^2 \tilde{C}} = 0.$$

By substituting our disease-free equilibrium value of B_v, we obtain

$$\mu_v N_v^* + \frac{\mu_v^2 N_v^*}{\theta^2} - \mu_v \tilde{n}_v N_v^* = \frac{\mu_v^2 \tilde{n}_v^2 (N_v^*)^2}{\theta^2 \tilde{C}}.$$

So it follows, by taking reciprocals that

$$\tilde{C} = \frac{\mu_v^2 \tilde{n}_v^2 (N_v^*)^2}{\theta^2 \left(\mu_v N_v^* + \frac{\mu_v N_v^*}{\theta^2} - \mu_v \tilde{n}_v N_v^*\right)}.$$

This can be simplified to

$$\tilde{C} = \frac{\mu_v \tilde{n}_v^2 N_v^*}{\theta^2 (1 - \tilde{n}_v) + \mu_v}. \tag{4}$$

The function for $\tilde{C}$ and the corresponding larval population size at this equilibrium is given in (3) and (4). We plot $\tilde{C}$ against $\tilde{n}_v \in [0, 1]$. This is shown below:

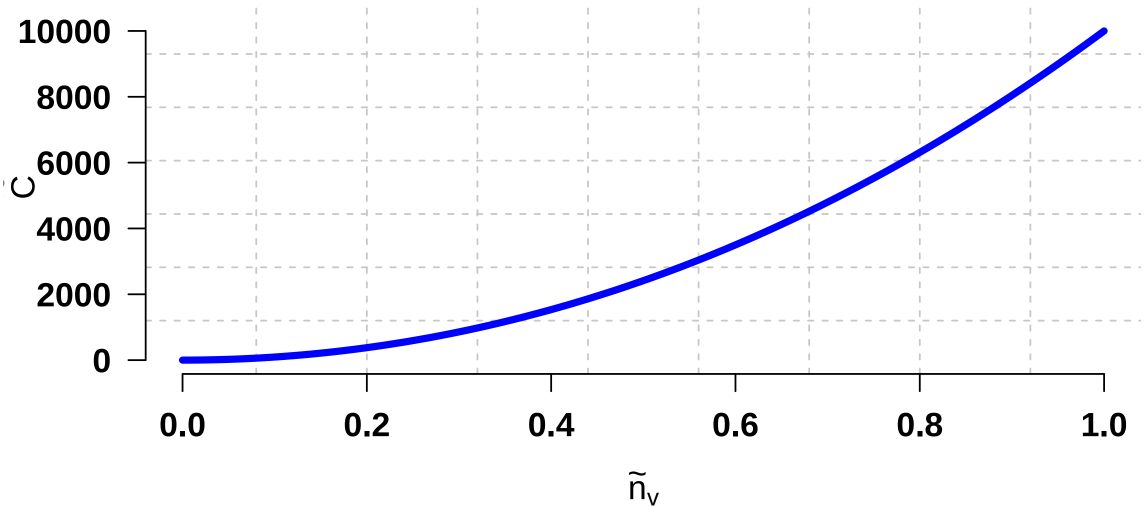


Figure 1: Plot of $\tilde{C}$ against $\tilde{n}_v \in [0, 1]$.

Q2a: To compute the within-farm R_0 for Farm 1, we adopt the Next Generation Matrix (NGM) approach. Firstly, we identify the states that are infected as being: E_h, I_h, Q_v, E_v and I_v. Next, we construct the differential equations for these compartments. Following the hint, we assume $K_{ij} = 0$ except for $K_{11} = 1$. The rate of influx of newly infected individuals for each of the aforementioned compartments is given by

$$\left[\frac{ap_h S_h K_{11} I_v}{N_h},\ 0,\ \frac{B_v q I_v}{N_v},\ \frac{ap_v S_v K_{11} I_h}{N_h},\ 0\right], \tag{5}$$

respectively. The net transfer of individuals out of each compartment is given by:

$$-\left(\left[(\sigma_h + \mu_h)E_h,\ (\gamma_h + \mu_h)I_h - \sigma_h E_h,\ \theta Q_v + \frac{Q_v(P_v + Q_v)}{C},\ (\sigma_v + \mu_v)E_v,\ \mu_v I_v - \sigma_v E_v - \theta Q_v\right]\right). \tag{6}$$

Now, we compute the Jacobian of these two representative systems in (5) and (6). The following matrix F will represent the rate of new infections from each class, specifically how susceptible individuals (both livestock and mosquitoes) become infected through the other compartments. The matrix V represents the rate at which individuals transition between compartments.

$$F = \begin{pmatrix} 0 & 0 & 0 & 0 & \frac{ap_h S_h K_{11}}{N_h} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{B_v q}{N_v} \\ 0 & \frac{ap_h S_v K_{11}}{N_h} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = -\begin{pmatrix} \sigma_h + \mu_h & 0 & 0 & 0 & 0 \\ -\sigma_h & \gamma_h + \mu_h & 0 & 0 & 0 \\ 0 & 0 & \theta + \frac{P_v + 2Q_v}{C} & 0 & 0 \\ 0 & 0 & 0 & \sigma_v + \mu_v & 0 \\ 0 & 0 & -\theta & -\sigma_v & \mu_v \end{pmatrix}.$$

The NGM is given by NGM $= -F \cdot V^{-1}$. The basic reproduction number, R_0, is then the spectral radius (dominant eigenvalue) of the NGM. We make use of Q1 to evaluate the system at the disease-free equilibrium. After inputting these matrices into R, we conclude that the within-farm R_0 is approximately `1.81`.

Q2b: We provide a plot of the spartial kernel K, and another plot of the location of the three farms, with the strength of the connections between farms.

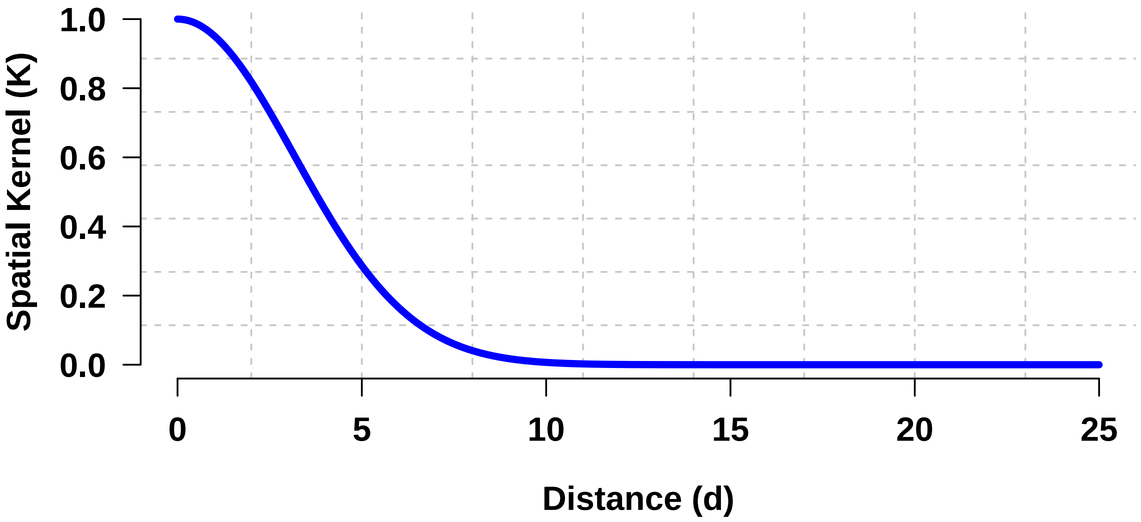


Figure 2: Plot of the spatial kernel as a function of distance $d \in [0, 25]$.

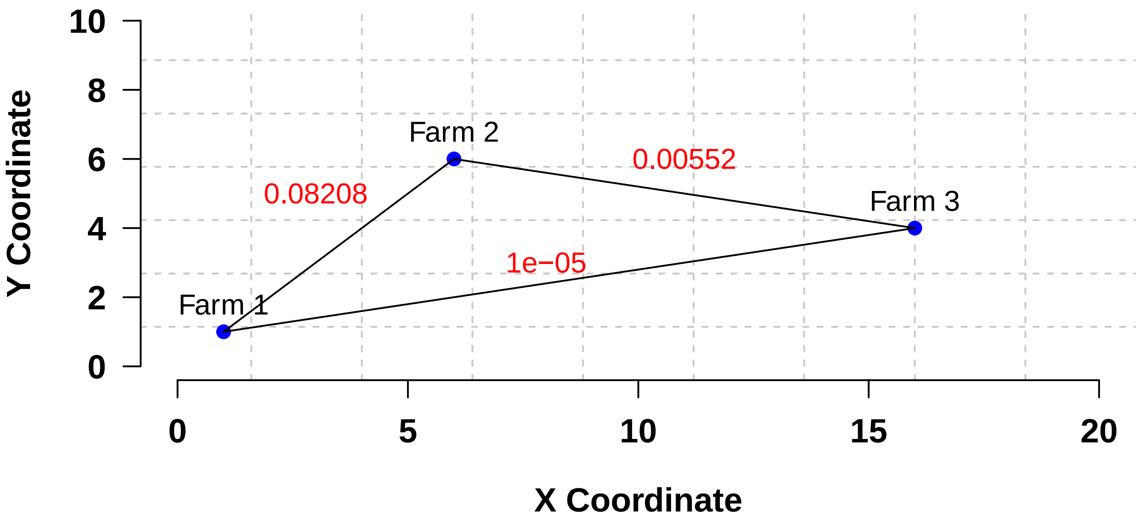


Figure 3: Plot of the locations of each farm and their respective pairwise spatial kernel strength, $K(i, j)$, rounded to 5 significant figures.

Q2c: After running the model with no controls (NC), we have the following infectious dynamics between hosts and vectors:

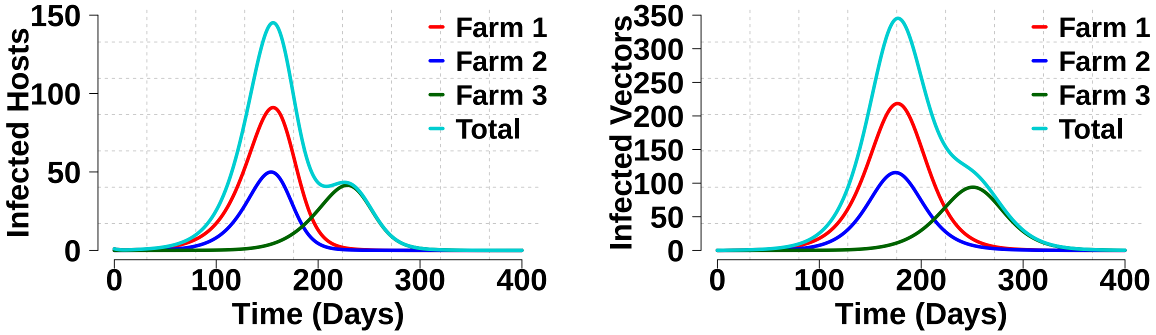


Figure 4: Plot of the infection dynamics for hosts (left) and vectors (right) over time.

In our model, vertical vectorial transmission refers to the direct transmission of infection from infected adult vectors I_v to their offspring, with the proportion of infected larvae governed by the parameter q. Increasing q results in more infected larvae, leading to a larger infected vector population. Consequently, the magnitude of the infected hosts also increases, but the overall shape of the outbreak dynamics remains the same. In model terms, a higher q increases the transmission from the non-infected larvae class P_v to the infected larvae class Q_v, causing a cascade effect that amplifies the number of infected hosts and vectors.

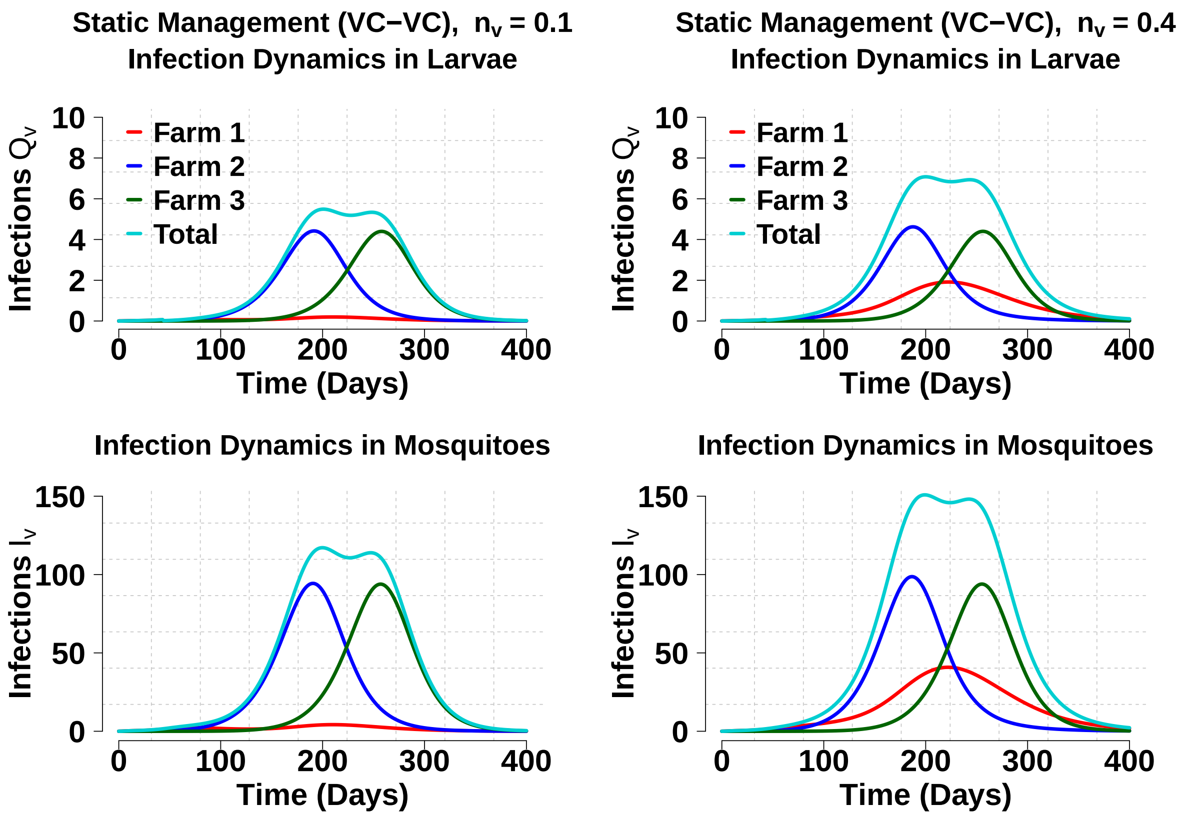
Q2d: The following table describes the duration and cost of this outbreak if each direct death of an animal costs \$250 and each failed pregnancy (abortion) costs \$50.

Strategy	Cost (\$)	Duration (Days)
NC	151,550	304

Table 1: Total cost and duration for the NC model.

The duration is calculated as the start up until when the outbreak is *over*, which is defined as each farm having < 1 infected (E_h or I_h) animal remaining.

Q3a: After running our model using the VC interventions for relative ATDs of $\tilde{n}_v = 0.1$ and $\tilde{n}_v = 0.4$, we obtain the following dynamics in the hosts, adult vectors and larval vectors.



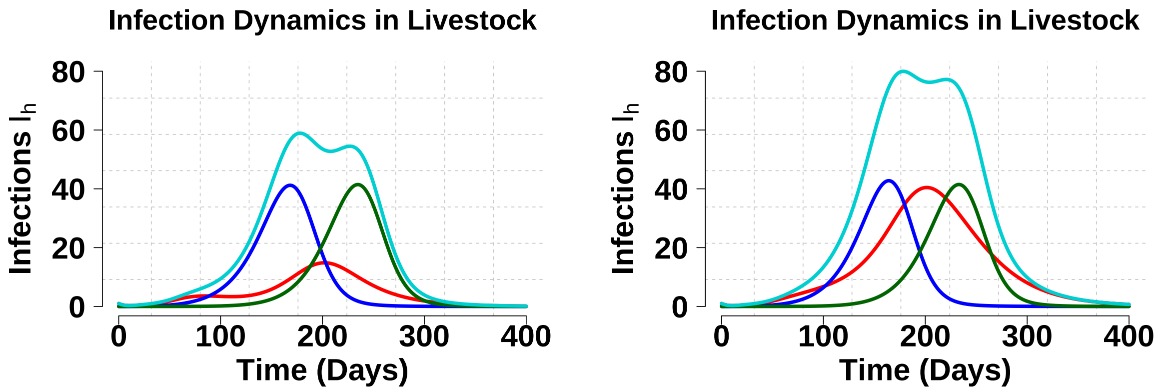


Figure 5: Infection dynamics for hosts, adult vectors (mosquitoes) and larval vectors, for static VC intervention (VC-VC).

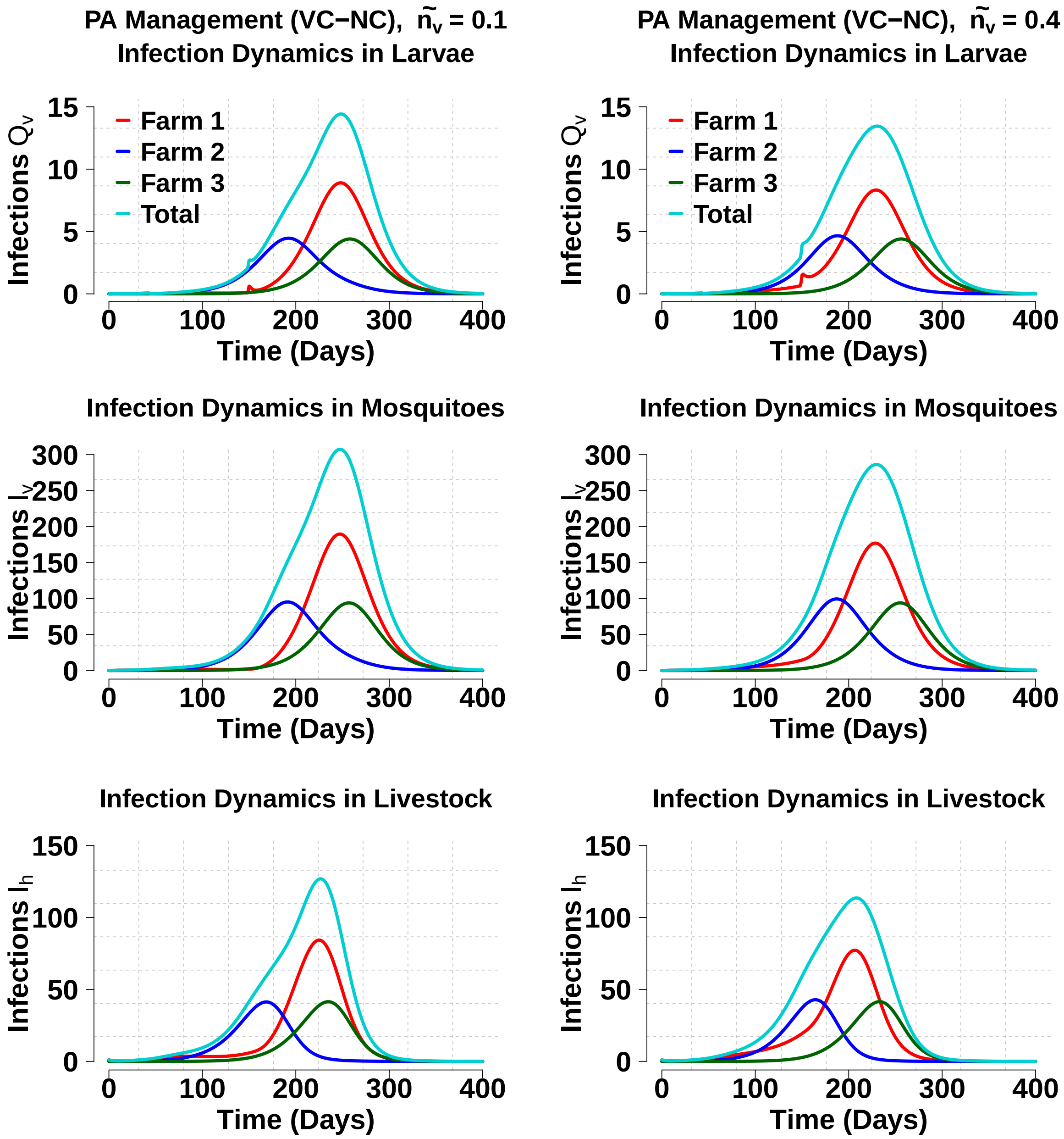


Figure 6: Infection dynamics for hosts, adult vectors (mosquitoes) and larval vectors, for passive adaptive management VC intervention (VC-NC).

	NC	**VC-VC (0.1)**	**VC-VC (0.4)**	**VC-NC (0.1)**	**VC-NC (0.4)**
Duration (Days)	304	323	385	311	309
Cost (\$)	151,550	117,700	166,450	158,000	158,450

Table 2: Summary of total cost and duration for different control strategies.

In the above table, NC refers to no control while VC-VC refers to a static intervention whereby we apply the vector control 7 days after outbreak detection until the end of the outbreak. VC-NC refers to a passive adaptive management intervention whereby we apply vector control 7 days after outbreak detection and it only lasts for 15 weeks before returning to no control.

The table above shows that the NC strategy performs better in terms of cost and duration compared to the VC-VC static at $\tilde{n}_v = 0.4$ as well as both of our passive adaptive management interventions. The only strategy that is potentially better with respect to our fundamental objectives (cost and duration) is the VC-VC static at $\tilde{n}_v = 0.1$, because although the duration an 8.2% increase, the reduction in cost of implementing the strategy is 22.4% which suggests there could be a trade-off to be made based on reducing cost and increasing duration. This depends on finding an optimal $\tilde{n}_v$ value.

Q3b: Lets assume that the relative ATD after vector control is approximately $\tilde{n}_v = 0.3$, with a distribution of our uncertainty given by $\tilde{n}_v^* \sim \text{Beta}(2,5)$. Then the expected duration and cost under the VC-VC and VC-NC strategies is given by:

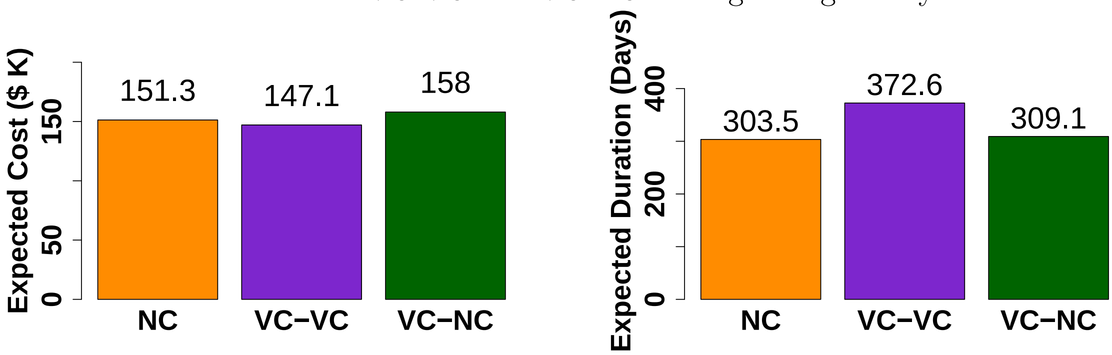


Figure 7: Expected cost and expected duration for NC, VC-VC and VC-NC strategies from the $\tilde{n}_v^* \sim \text{Beta}(2,5)$ distribution.

Assuming our fundamental objectives are both duration and cost, we wish to minimise both variables. The first observation is that the NC strategy costs less than the VC-NC strategy and has a lower expected duration suggesting irrespective of our fundamental objective being minimising cost or minimising duration, we always choose NC over VC-NC.

If our fundamental objective is to minimise cost, then we should choose strategy VC-VC as this incurs the least cost.

If our fundamental objective is to minimise duration, then we should choose strategy NC

as this has the least expected duration.
Based on this analysis, we rank the VC-NC strategy as the worst and depending on our fundamental objective we determine the order of the remaining two strategies. If we wish to account for both fundamental objectives, and we assume a linear distribution of cost across each day, then we observe that the VC-VC strategy has 69.1 additional days in duration. By choosing the NC strategy over the VC-VC strategy, we essentially value 69.1 days of outbreak at a value of the difference in costs between strategies which is $\$2,201$. By linearly extrapolating the NC strategy by 69.1 days, it we would save $(151294/303.5) * 69.1 \approx \$34,446$. This suggests that by taking both fundamental objectives into account and assuming a linear distribution, that we should choose the NC strategy. In this case, we rank the strategies as NC, VC-VC and VC-NC.

Q3c: We employ an active adaptive management strategy where we assume that if we monitor ATD, we gain perfect information. The following plots show how duration and costs change when we vary $\tilde{n}_v \in [0, 1]$.

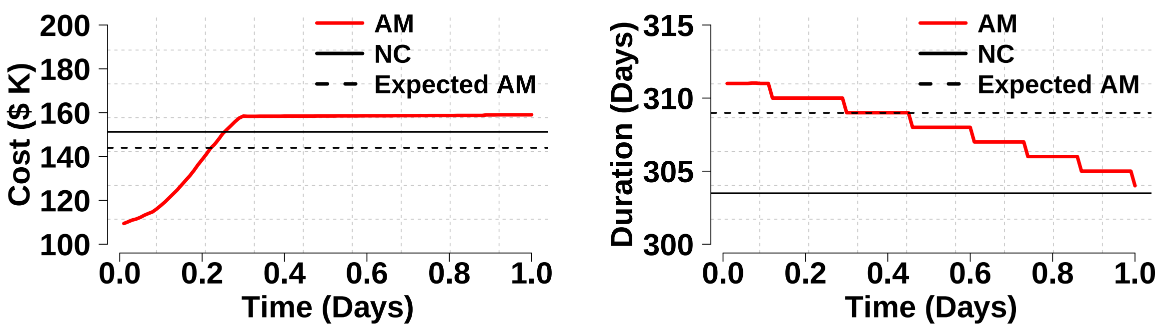


Figure 8: Plot of active adaptive management (AM) against NC across values for $\tilde{n}_v \in [0, 1]$.

From this, we conclude that the recommendation is to not do active AM if the objective is to minimise duration as NC has a lower duration than AM for all values of $\tilde{n}_v$. However, should the objective be to minimise cost, then we should employ VC initially and then assess at time t^* and switch to NC.

Q4a: Alternative interventions against RVFV could include:

- Applying vaccination to livestock which reduces the susceptibility class by creating another vaccinated class. This would involve introducing parameters for the proportion of livestock vaccinated and the success rate of the vaccination.

- Quarantining and restricting the movement of livestock to prevent the spread of infection between infected livestock. This would include adding another term to the infected classes which control the movement of livestock.

- Insecticides or sprays which reduce larvae populations which would involve introducing a negative term to the larvae class. This would have a knock on reduction effect into the mosquitoes class.

- There could be an intervention of culling exposed livestock which would involve adding a penalty term to the exposed class. Costs would also have to be considered here and this would be an ideal intervention if our fundamental objective was duration and not cost.

If our fundamental objective is duration, then implementing interventions early would be most ideal as this would immediately reduce the spread of infection. However, the suggested interventions for doing this would amount to increased costs. We could delay the timing of our interventions to better understand the trade-off between immediate action and better-targeted interventions.

Other than duration and costs, other policy maker objectives that could be considered include: minimising animal suffering by reducing morality and morbidity in livestock; ensuring food security by protecting and maintaining the livestock population, in this case vector control interventions are better suited.

Mosquito populations are heavily influenced by the weather therefore, having data about weather conditions in the regions where the farms are located could prove useful for deciding strategies. If the weather is heavily favourable for mosquitoes, then vector control interventions may be much more suited. Conversely, if the weather is already limiting the production of mosquitoes from larvae, then using a livestock controlling intervention may be more cost-effective. Additionally, if we had more data about trade and movement of livestock between farms, we could better model this instead of using distance as our only factor. This would improve our model in understanding transmission of infection between farms which could give us better insight as to whether livestock movement contributes more to the spread of infection. If so, then we could consider adding livestock quarantining interventions as mentioned earlier.

Q4b: Three modifications which we could make to our model to make it more realistic and practical for guiding RVFV policy include:

- Include seasonal and environmental variability because our current model assumes constant birth, death and infection rates but realistically, these are influenced by the season.
- Since the model situates around a farm, humans are included in the system so by including human spillover transmission into the model would make it more realistic. This would make the model more suitable for a public health policy about RVFV.
- Include stochasticity to reflect randomness which is present in the real world. Practically, there is always chance involved in such a system and by including this into the model would make it more realistic.